

Power Plant Emission Minimization

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Abstract - This work presents a surrogate-based optimization approach for reducing emissions in a conventional steam-cycle power plant using historical operating data. Regression models were developed to relate six selected operating variables to five emissions outputs and three plant performance metrics, allowing the optimization problem to be evaluated efficiently without repeated high-fidelity simulation. A weighted normalized emissions objective was formulated and minimized subject to constraints on net plant heat rate, net turbine heat rate, higher heating value, and feasible operating bounds. Multiple optimization strategies were examined, including steepest descent, conjugate gradient, particle swarm optimization, and sequential quadratic programming, with particle swarm optimization yielding the best feasible result among the comparable linear-surrogate methods. Sensitivity analysis was then used to identify the most influential operating variables and active constraints near the optimum. Finally, the problem was extended to a bi-objective formulation in which emissions and net plant heat rate were considered simultaneously, and a Pareto front was generated using the Normal Boundary Intersection method to characterize the tradeoff between environmental performance and thermal efficiency.

I. INTRODUCTION

In a conventional steam-cycle power plant, coal combustion in the boiler produces high-temperature flue gas and steam, the steam expands through the turbine-generator to produce electricity, and the flue gas passes through cleanup equipment before being released through the stack. Although these plants are usually operated with strong emphasis on power output and heat-rate performance, the same combustion and gas-handling process also generate pollutants such as SO₂, NO_x, CO, CO₂, and dust, making emissions an essential design consideration rather than a secondary one [1]. In this work, plant operating data collected over 2022 at 10-minute intervals is used to build regression-based surrogate models that relate a selected set of operating variables to both emissions and efficiency metrics, enabling fast optimization without repeatedly solving a high-fidelity plant model[6].

II. SYSTEM ARCHITECTURE

The coal-fired steam power plant was decomposed into seven interacting modules - Coal Feed/Fuel Quality, Air and Draft System, Boiler and Combustion, Steam-Water Cycle, Turbine Generator, Cleanup, and Stack; to capture the physical flow of fuel, air, steam, and flue gas through the system. Based on this decomposition, an N2 matrix was developed by identifying the major inputs and outputs of each module and marking each instance in which the output of one subsystem served as an input or feedback signal to another. The resulting matrix highlights the strong couplings among fuel quality, air handling, combustion, steam generation, turbine operation, and emissions cleanup, and shows that both efficiency and emissions are governed by interactions across multiple subsystems rather than by any single component in isolation.

Coal Feed Fuel quality		1 (HHV)				
	Air & Draft System	1 (APH Leakage)			1(APH Leakage)	1(APH Leakage)
		Boiler & Combustion	1(Temp; Energy input from Boiler)		1 (Creates Pollutants) (SO ₂ ; NO _x ;	
		1 (Feed Water flow, Feed water temperature)	Steam Water Cycle	1 (Feed water flow; Energy input)		
				Turbine Generator		
					Clean up	(SO ₂ , NO _x , Dust)
	(SO ₂ , NO _x , Dust)	(SO ₂ , NO _x , Dust)			(SO ₂ , NO _x , Dust)	Stack

Figure 1. N2 Matrix

The block diagram provides a complementary view of the same architecture by illustrating the direction and nature of these interactions. In the diagram, solid lines denote explicit inputs, dotted lines denote implicit inter-module coupling effects, and dashed lines denote feedback signals used for operational control. This representation shows that the plant operates as a closed-loop system in which, for example, the Air and Draft System respond to feedback from the Stack and Turbine Generator, while the Boiler and Combustion subsystem adjusts to turbine-side demand signals. Together, the N2

matrix and block diagram establish the multidisciplinary basis of the problem formulation and justify the use of surrogate-based optimization to study how changes in operating variables propagate through the plant to affect emissions and performance.

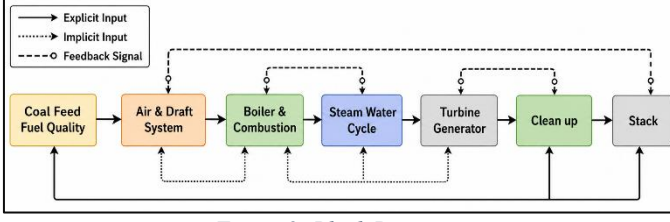


Figure 2- Block Diagram

III. PROBLEM FORMULATION

The goal of this project was to determine a feasible operating point for the power plant that minimizes a composite emissions objective while satisfying efficiency-related requirements and practical operating bounds. The original dataset contained 55 operational parameters. However, some of the parameters were recorded for monitoring purposes only. Thus, to find the varying parameters, that could be used for optimization of the power plant, the percent change of standard deviation by mean was calculated. The parameters with percent change higher than 10% were kept for optimization. Using that screening process, the final formulation keeps six design variables: superheater desuperheating water flow x_1 , reheater desuperheating water flow x_2 , APH leakage x_3 , boiler energy input x_4 , feedwater inlet temperature x_5 , and feedwater mass flow rate x_6 .

A. Mathematical Model

The plant is modeled with regression-based surrogate equations for five emissions (SO_2 , NO_x , CO , CO_2 , Dust) and three performance constraints (NPHR, NTHR, HHV), all expressed as functions of the six design variables. In the system architecture used for the project, the plant is decomposed into interacting modules [Coal Feed/Fuel Quality, Air Draft System, Boiler Combustion, Steam Water Cycle, Turbine Generator, Cleanup, and Stack] to represent the physical couplings that justify a multidisciplinary formulation. This modeling choice allows the optimization to capture how changes in combustion-side and steam-cycle variables propagate to both emissions and efficiency measures across the plant.

The scalar objective is a weighted sum of normalized emissions,

$$J(x) = 0.25\widetilde{\text{SO}}_2(x) + 0.25\widetilde{\text{NO}}_x(x) + 0.25\widetilde{\text{CO}}(x) + 0.15\widetilde{\text{CO}}_2(x) + 0.10\widetilde{\text{Dust}}(x)$$

This weighting structure was adopted so that toxic gas emissions receive greater emphasis, while normalization

prevents large-magnitude outputs such as CO_2 from dominating the objective purely because of scale.

For the linear surrogate formulation used in the gradient-based and swarm-based comparisons, the expanded objective is reported as approximately

$$J(x) \approx 0.001 - 0.0686 x_1 + 0.2146 x_2 - 0.3432 x_3 + 0.0 x_4 + 0.4996 x_5 + 0.4639 x_6.$$

The later comparison report notes that the coefficient of x_4 is effectively zero in the emission objective, but x_4 remains essential because it appears strongly in the constraint surrogates and is necessary for feasibility, especially for satisfying the HHV requirement.

The optimization problem is therefore written as

$$\min_x J(x)$$

subject to

$$\begin{aligned} \text{NPHR}(x) &\leq 2750, \\ \text{NTHR}(x) &\leq 2600, \\ \text{HHV}(x) &\geq 4240, \end{aligned}$$

along with lower and upper bounds on all six design variables taken from the observed plant operating envelope.

IV. OPTIMIZATION

The optimization stage of this project was formulated using the corrected six-variable problem statement developed from the plant data and surrogate models. For each candidate design vector $x = [x_1, x_2, x_3, x_4, x_5, x_6]^T$, regression-based surrogate models were used to predict the five emission outputs SO_2 , NO_x , CO , CO_2 , and Dust , as well as the performance metrics NPHR, NTHR, and HHV[1][6].

The main objective was to minimize the weighted normalized emission function, where each emission term was normalized by its dataset mean so that no single quantity dominated the optimization solely because of its magnitude [2].

The design variables were bounded by the historical operating envelope of the plant [6]: $65 \leq x_1 \leq 95$, $1 \leq x_2 \leq 5$, $0 \leq x_3 \leq 4$, $695,000 \leq x_4 \leq 725,000$, $225 \leq x_5 \leq 245$, and $815 \leq x_6 \leq 855$. In addition, the optimization was constrained by $\text{NPHR}(x) \leq 2750$, $\text{NTHR}(x) \leq 2600$, and $\text{HHV}(x) \geq 4240$, which ensured that the search did not reduce emissions by moving to unrealistic or thermally unacceptable operating conditions.

A. Constraint Handling

To enable a consistent comparison among algorithms, constrained optimization was handled through penalization. A quadratic external penalty was added to the objective for violations of the performance constraints,

and an additional strong penalty was applied to box-bound violations where needed, so that infeasible candidate solutions were disfavored without changing the original engineering interpretation of the constraints.

The corrected coefficients show that x_4 contributes substantially to the HHV surrogate, and at the upper bound $x_4 = 725,000$ its contribution is approximately 1980 kcal/kg, which is critical to satisfying the lower limit of 4240 kcal/kg. As a result, the optimal region lies near active constraints, and optimizer performance depends strongly on the balance between descent efficiency and reliable feasibility preservation.

B. Design of Experiments

Since the data was continuous with over 50,000 data points, the range of each design variable was divided into three intervals: low, middle, and high; the mean of each interval was computed and used as the representative value for that level. The One-at-a-Time method was selected as it provides an efficient means of identifying a suitable starting point with minimal computational cost. In the first iteration, all parameters were set to their middle level to capture the effect of both increasing and decreasing each parameter relative to its nominal value. After 13 iterations, the optimal starting point was identified, as summarized in the table below.

Level Symbol	Description	Range in the Best Combination	Mean Value of the Range in the Best Combination
A	Desuperheating water flow to superheater (t/h)	Middle	65
B	Desuperheating water flow to reheater (t/h)	High	1
C	APH leakage (%)	Middle	0
D	Energy Input from Boiler (kcal/h)	Low	695000
E	Feedwater inlet temperature to boiler (°F)	Low	225
F	Feedwater mass flow rate (t/h)	Middle	815

Table 1: Results of DOE, Best Starting Combination

C. Algorithms Evaluated

Four algorithms were considered in the refined study: steepest descent, conjugate gradient, particle swarm optimization, and sequential quadratic programming. Steepest descent with Armijo backtracking was used as the

baseline gradient-based method because the linear surrogate formulation provides simple analytical gradients, but in practice it made little progress because the normalized objective had a very small gradient magnitude, causing the algorithm to accept a negligible step and terminate near the initial design with $J(x) = 0.4992$. To improve first-order performance, conjugate gradient was applied to the same corrected six-variable problem after normalizing all variables to $[0, 1]$, which removed the large-scale mismatch between boiler energy input and the remaining design variables and enabled meaningful descent, reducing the objective to $J(x) = 0.4855$. Particle swarm optimization was included as a global, derivative-free benchmark and proved well suited to the affine linear-surrogate objective, converging to a boundary solution with the best comparable result, $J(x) = 0.4839$. The Sequential quadratic programming solution satisfied all constraints as well, but because it used quadratic surrogates its normalized objective value of 0.9825 should be interpreted separately rather than as a direct ranking against the linear-model methods.

D. Conjugate Gradient Solution

The conjugate gradient (CG) method was applied to the corrected six-variable linear-surrogate formulation as the primary improved gradient-based optimization approach. The optimization problem was solved using the Fletcher-Reeves implementation, where the objective was the weighted normalized emission function and feasibility was enforced through quadratic penalties on the NPHR, NTHR, and HHV constraints. Additional penalties were used to enforce the variable bounds, ensuring that all candidate solutions remained within the historical operating envelope of the plant.

A key modification required for successful CG performance was the normalization of all design variables to the interval $[0, 1]$. This step was necessary because the boiler energy variable x_4 has a magnitude on the order of 10^5 , whereas the remaining variables range between approximately 1 and 855, and such a scale disparity can distort the search direction and degrade numerical conditioning. By scaling the variables before optimization, the algorithm was able to respond to actual sensitivity in the objective function rather than to raw differences in variable magnitude.

The CG method was initialized at the midpoint of the feasible design space, $x_0 = [80, 3, 2, 710000, 235, 835]$. Compared with steepest descent, conjugate gradient provided more effective progress because each new search direction incorporated information from previous iterations rather than following only the local negative gradient. This resulted in a lower feasible objective value of $J(x) = 0.4855$, compared with $J(x) = 0.4992$ for the corrected steepest descent implementation.

The final CG solution was $x_1 = 88.70$, $x_2 = 2.78$, $x_3 = 2.50$, $x_4 = 725,000$, $x_5 = 225.00$, and $x_6 = 817.91$. At this operating point, the predicted constraint values were $\text{NPHR} = 2699.9$, $\text{NTHR} = 2529.2$, and $\text{HHV} = 4240.0$, indicating that the solution satisfied all feasibility requirements while lying close to the active HHV constraint boundary. The corresponding emission predictions from the linear surrogate were $\text{SO}_2 = 234.87$, $\text{NO}_x = 93.36$, $\text{CO} = 129.21$, $\text{CO}_2 = 51,523$, and $\text{Dust} = 4.07$.

Although the solver reported a precision-loss warning and terminated after 25 iterations without full numerical convergence, the final design remained feasible and clearly improved upon the steepest descent baseline. These results indicate that conjugate gradient was the best-performing gradient-based method in the study, primarily because variable normalization allowed the method to exploit conjugate search directions effectively in a problem with strong scaling differences. However, the slightly lower objective obtained by PSO suggests that CG approached, but did not fully reach, the best corner solution of the linear-surrogate feasible region.

E. Particle Swarm Optimization

Particle Swarm was chosen as one of the main heuristic algorithms due to multiple reasons. Firstly, the heuristics ability to handle discrete or mixed variables is perfect for this application. Secondly, since the CG method was done as well, the group also wanted to ensure a globally optimal solution was found by comparing this to the other result. Lastly, PSO is not too computationally expensive for the machines that the group would be running.

F. PSO Implementation and Methodology

PSO was implemented using a penalty method to properly handle the mixed types of constraints[4]. As stated in the previous section, the three constraints of the Particle Swarm are Net Plant Heat Rate (NPHR), Net Turbine Heat Rate (NTHR), and the Higher Heating Value (HHV). The three constraint functions are shown below:

$$G_1(x) = 0.2598x_1 + 0.3361x_2 + 0.2394x_3 + 0.00000173x_4 + 1.1498x_5 + 1.4238x_6 - 2750 \leq 0$$

$$G_2(x) = 0.2379x_1 + 0.2909x_2 + 0.3460x_3 + 0.00000164x_4 + 1.0487x_5 + 1.3266x_6 - 2600 \leq 0$$

$$G_3(x) = 4240 - (0.3817x_1 + 0.3432x_2 + 0.2709x_3 + 0.000002731x_4 + 1.7837x_5 + 2.2291x_6) \leq 0$$

The three constraint functions ensure the values stay within their limits and overall minimize $J(x)$. The way PSO functions, is in four main components:

- **Particles:** A solution with a position x (current design) and velocity v (direction and magnitude of movement)
- **Personal Best:** The overall best position found by each particle
- **Global Best:** The best position found by the while swarm
- **Social Learning:** How the particles learn from their previous experience and the swarms experience

G. Algorithm Parameters and Reasoning

The algorithm parameters for the Python function are listed below in the table:

Parameter	Value
Swarm Size	120
Max Iterations	300
Function Tolerance	1E-8
Inertia Weight	0.7
Cognitive	1.5
Social	1.5

Table 2: Algorithm Parameters

For swarm size, it is typically good practice to have the number of variables multiplied by 20 as a good baseline. The initial starting point was 50 particles, but this had issues occasionally not reaching a global optimum and getting stuck at a local point. 300 max iterations were selected because it was sufficient for consistent convergence. For the function tolerance, 1E-8 was selected to have a high-fidelity answer for our final answer. Inertial weight, this value stays constant at .7, where this is a good balance between exploration and exploitation. The cognitive particle parameter was selected to stay at the default value of 1.5 set by the user. This also goes for the social parameter that affects how the particles communicate with each other.

The initial strategy was to have 10 percent of the swarm start at the point gained from the Design of Experiments analysis. The other 90% of the particles are uniformly distributed within bounds using a Latin Hypercube-style random initialization, as this is the default with the integrated NumPy feature in Python.

The penalty parameter is 1E6 which was chosen for two reasons. One, a large penalty parameter creates a function that has firm boundaries for the constraints. Two, the possible infeasible solutions get a large penalty which constantly drives the swarm to a feasible solution. Too low of a penalty function may converge on an infeasible solution and will take more time for convergence. A penalty function too high will have some instability issues when converging.

H. PSO Results

x1 (SH Spray Flow) (t/h)	95
x2 (RH Spray Flow) (t/h)	1
x3 (APH Leakage) (%)	4
x4 (Boiler Energy Input) (Mcal/h)	725,000
x5 (Feedwater Temp) (°F)	225
x6 (Feedwater Flow) (t/h)	817.251289

Table 3: PSO Results

For all of the design variables except for x6, are at their respective bounds. The respective constraint function equations and their output are shown below:

Constraint	Metric	Limit	Margin	Status
G1	2702.53	≤ 2750	47.47	Good
G2	2533.40	≤ 2600	66.60	Good
G3	4240.01	≥ 4240	0.006	Good

The final iteration of this method was 11 where the $f = .48385767$ with the total number evaluations is 1320 with 120 particle for each iteration. This fast convergence was due to a few reasons:

- Good initial guess of x0 from DOE
- Linear objective
- Simple design space: The constraints create a convex feasible region

V. SENSITIVITY ANALYSIS

A. Method

To understand how significantly the objective function was affected by changes in the constraints and parameters, sensitivity analysis was performed at the PSO optimal point x^* . Since a linear function for the objective had been obtained, the gradients of the objective function with respect to each variable could be derived analytically. To validate the analytical gradients, a numerical sensitivity analysis was also performed using the central finite difference method, which was selected for its second order accuracy, which was more accurate than both the forward and the backward difference methods. The step size was set at 10^{-5} , which was validated in a step size analysis. The maximum error between the analytical and numerical methods was in the order of $e-12$, which is about the precision of computer calculations. This confirms that both analytical and numerical methods agree. Sensitivity analysis was also performed with respect to the fixed parameters of the problem, specifically the constraint limits and the emission weights of the weighted objective function. To avoid redundancy, the rest of the analysis was done using the numerical method only.

B. Results

Figure 3 shows that the parameter with the highest sensitivity is feedwater flow with 3.84%. On the other hand, even though boiler energy has a significant effect on the feasibility of the solution, it exhibits zero sensitivity regarding the objective function, which means boiler energy does not affect the objective function value directly.

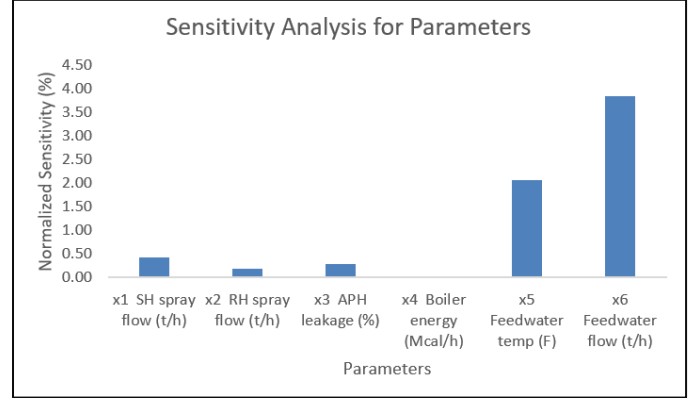


Figure 3. Sensitivity Analysis Results for Parameters

Doing the sensitivity analysis for constraints, it was observed that the only active constraint was HHV. Since there was no need to compare results, the sensitivity of the penalized objective with respect to constraint limits was not normalized. By increasing the HHV constraint value by 1Kcal/Kg, the penalty function would increase by 7.225. Also, it was observed that the step size 10^{-5} was too small to capture the sensitivity of constraints, therefore, the analysis was repeated using a larger step size, 1.

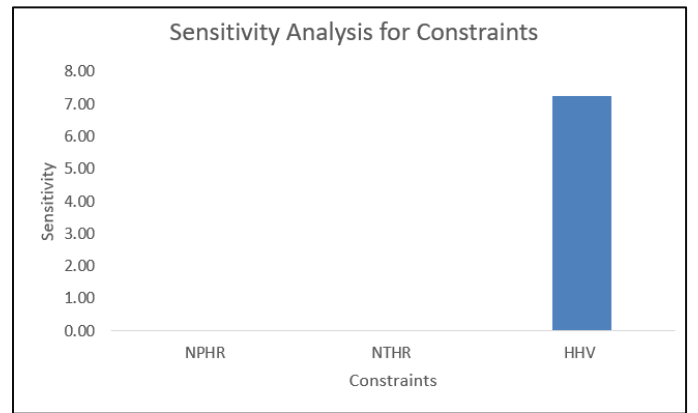


Figure 4. Sensitivity Analysis Results for Constraints

For the weights of emissions in the objective function, it was observed that the weights of SO_2 , NO_x , and CO, were the most influential, 1% change in each caused

approximately 0.51% change in the objective function. Dust had the smallest sensitivity with 0.21%.

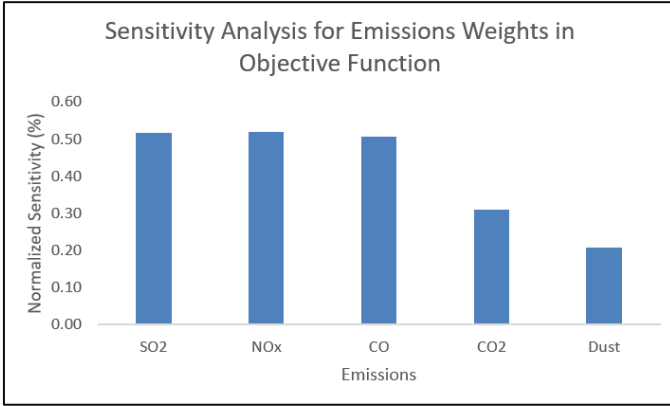


Figure 5. Sensitivity Analysis Results for Emissions Weights.

C. Sensitivity Analysis at CG Optimal Point

Sensitivity analysis was also performed at the optimal point obtained from the CG method. The normalized sensitivities for the design variables and emission weights were very similar to those from the PSO optimal point. Regarding the constraints, similar to PSO, only the HHV constraint was active; however, since the CG optimal point was almost on the HHV bound, the sensitivity was significantly higher at 100.23, compared to 7.255 for PSO.

VI. MULTI-OBJECTIVE OPTIMIZATION

In practical plant operation, reducing emissions and improving efficiency are often competing goals, since operating changes that improve one may worsen the other. Hence, the problem was extended to a bi-objective framework in which emissions and thermal performance were considered simultaneously [3]. This extension provides a more complete view of the feasible design space and allows compromise operating points to be identified rather than manipulating the analysis to return only one preferred solution.

A. Bi-objective Optimization

To construct the multi-objective formulation, the original scalar emissions objective was retained as the first objective and the net plant heat rate (NPHR) was introduced as the second objective. The first objective, $J_1(x)$, represents the weighted normalized sum of the predicted emissions of SO_2 , NO_x , CO , CO_2 , and dust, while the second objective, $J_2(x)$, represents NPHR as a measure of plant thermal performance. Because a lower NPHR corresponds to better efficiency, both objectives were posed as minimization functions.

The constraint set associated with $J_1(x)$, was preserved from the original formulation. In the bi-objective case, NTHR and HHV were retained as performance constraints, and additional gross load and net load constraints were introduced to prevent optimization from selecting low-emission solutions that would compromise practical power output.

$$J_1(x) \approx 0.001 - 0.0686 x_1 + 0.2146 x_2 - 0.3432 x_3 + 0.0 x_4 + 0.4996 x_5 + 0.4639 x_6$$

$$J_2(x) \approx 0.00 + 0.2598 x_1 + 0.3361 x_2 - 0.2394 x_3 + 0.0 x_4 + 1.1498 x_5 + 1.4238 x_6$$

B. Pareto Front

To characterize the tradeoff between emissions and plant performance, the Pareto front was generated using the Normal Boundary Intersection (NBI) method. This method was selected because it provides a more evenly distributed set of Pareto-optimal points than a simple weighted-sum approach [3], particularly when the tradeoff curve is nonlinear or unevenly shaped. The resulting front consists of feasible non-dominated solutions, meaning that improvement in one objective can only be achieved at the expense of the other.

The NBI method was therefore used to trace the tradeoff boundary between the emissions objective and NPHR across the feasible operating region. This approach provides a clearer description of the available compromise solutions than a single-objective formulation alone.

C. Results

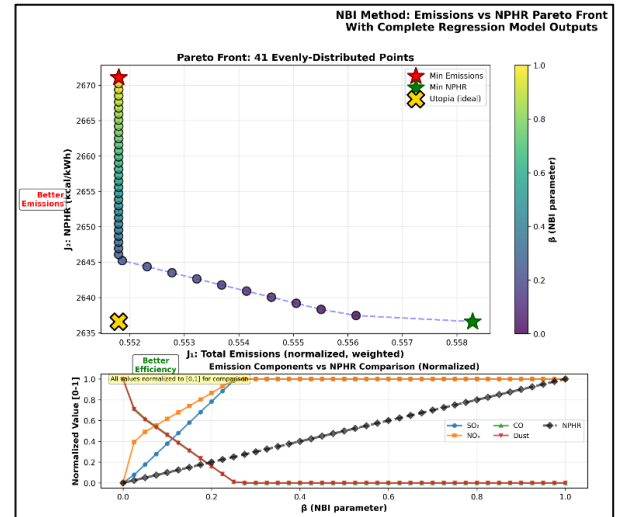


Fig 6. Pareto Front Analysis

The Pareto results showed that the two objectives are opposing. One end of the front corresponds to minimum

NPHR, which represents improved thermal efficiency but worse emissions performance, while the other end corresponds to minimum emissions with poorer heat-rate performance [4]. Points located between these extremes represent compromise operating conditions and are therefore of greatest practical interest, since they balance environmental impact and efficiency rather than favoring only one metric.

The Pareto front also showed that the preferred operating point depends on decision preference. Different parameter selections in the NBI procedure emphasized different regions of the front, shifting the solution toward either emissions reduction or efficiency improvement. In this sense, the Pareto front defines the boundary of the best available tradeoff solutions and provides a more informative basis for decision-making than a single-objective optimum alone.

This interpretation is consistent with the earlier single-objective and sensitivity results. The PSO-based solution achieved the lowest emissions objective value among the tested methods, approximately 0.4839, while remaining feasible under the imposed constraints, and the sensitivity analysis showed that feedwater temperature and feedwater flow rate were among the most influential variables, whereas boiler energy input had negligible local sensitivity near the optimum. In addition, HHV was identified as the active constraint, indicating that the Pareto shape is governed not only by the conflict between emissions and efficiency but also by the limited feasible operating region.

VII. FINAL DESIGN

The final design selected in this study corresponds to the feasible operating point obtained using Particle Swarm Optimization (PSO), since this method produced the lowest value of the single-objective emissions function among the optimization methods evaluated using the linear surrogate model. The PSO solution achieved a normalized objective value of $J(x)=0.4839$, which was lower than the values obtained using conjugate gradient and steepest descent.

The corresponding final design variables are $x_1=94.97$ t/h for superheater spray flow, $x_2=1.00$ t/h

for reheater spray flow, $x_3=4.00$ for APH leakage, $x_4=725000$ Mcal/h for boiler energy input, $x_5=225.00$ °F for feedwater inlet temperature, and $x_6=817.2512$ t/h for feedwater flow rate. This operating point was selected as the final design because it minimizes the weighted normalized emissions objective while satisfying the imposed NPHR, NTHR, and HHV constraints within the feasible plant operating region.

At this design, the predicted performance values are $NPHR = 2700.36$, $NTHR = 2529.77$, and $HHV = 4240.73$. These results confirm that the final design is feasible and lies close to the active HHV constraint, indicating that the minimum emissions is achieved near the boundary of the allowable operating space.

From the standpoint of single-objective optimization, this PSO solution is the most appropriate final design because the primary goal is emissions minimization subject to feasible plant operation. Although alternative tradeoff solutions are available in the multi-objective analysis, the PSO point remains the best design when emissions reduction is treated as the dominant design objective.

VIII. CONCLUSION

This project developed a surrogate-based optimization framework for minimizing emissions in a conventional steam-cycle power plant while maintaining feasible thermal and operational performance. Regression-based surrogate models were constructed from plant operating data and used to solve the optimization problem efficiently over a six-variable design space. Among the methods evaluated, Particle Swarm Optimization produced the best feasible single-objective result, while conjugate gradient showed the strongest performance among the gradient-based methods after variable normalization. The results also indicated that feedwater temperature and feedwater flow rate were among the most influential variables, whereas boiler energy input contributed primarily to feasibility through the HHV constraint rather than directly to the emissions objective.

The multi-objective extension showed that emissions reduction and thermal efficiency are conflicting goals in practical plant operation. The Pareto front generated

using the Normal Boundary Intersection method demonstrated that improvement in one objective can only be achieved at the expense of the other, so the preferred operating point depends on the desired balance between environmental and efficiency considerations. Overall, the study confirms that data-driven optimization provides a practical and effective approach for identifying improved plant operating conditions under realistic constraints.

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